Quest 11: The Portalignment

Lord of the Wings · shortcuts across a ring of portals

> This guide uses the official puzzle text. The story is summarised and the rules are restated in plain words. It explains what the quest asks and the logic behind my solutions, not the code itself. Every number in the examples and traces was checked by running my solutions on the sample data (plus small instrumented copies to print intermediate values).

THE QUESTION

Portals sit in a ring and are linked to their neighbours; extra two-way shortcuts cut across the ring. Add up the shortest distance between **every** pair of portals, and report how much the shortcuts reduce that total.

INPUT

`portals: N`, then one shortcut per line (`a b`).

OUTPUT

The *portal resonance* = strain without shortcuts − strain with them.

WHAT CHANGES BETWEEN PARTS

Part I links each portal to its 1 nearest neighbour on each side. Part II links the 3 nearest on each side and lets you add **one** extra connection anywhere. Part III uses the Part II ring and adds **four** extra connections, made by 8 crystals spaced evenly round the ring and paired up.

The story

Once the Nullghosts have faded, the team goes deeper under Nulldor and finds a huge round chamber with an ancient portal ring, the only way up into the fortress. Gimlock knows how to tune it: measure how much the known shortcuts shrink the total travel distance. Two more rings follow on the levels above, and for those the team can place tuning crystals to make extra shortcuts of their own, as cleverly as possible.

Understanding the question

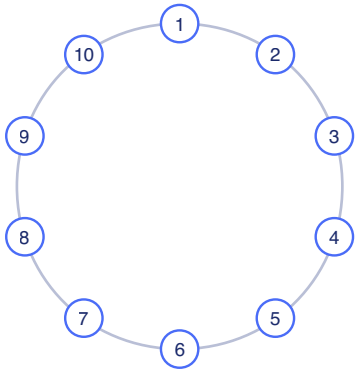
The rules

```
strain(graph) = Σ over every start portal s
                  Σ over every other portal t   (fewest teleports from s to t)

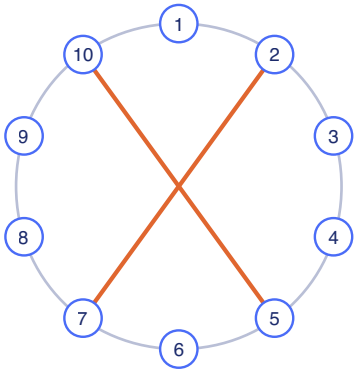
resonance = strain(ring only) − strain(ring + shortcuts [+ crystals])
```

- The distance between two portals is the **number of teleports** on the shortest route: every connection costs 1, whether it's a ring link or a long shortcut. In graph words: an unweighted graph, so **breadth-first search** (BFS) gives exact distances.
- The strain counts every ordered pair, so the trip 1 → 5 and the trip 5 → 1 are both included. That's why the plain 10-ring has strain 25 × 10 = 250, not 125.
- Shortcuts work both ways. A shortcut can never make the strain larger, only equal or smaller.

The sample ring (Parts I and II)



ring only: strain 250



with 5–10 and 2–7: strain 190

Extra examples of my own (run through copies of my Part I solution)

Notes	Resonance	Why
10 portals, `1 2`	0	1 and 2 are already neighbours, so the shortcut adds nothing.
10 portals, `1 6`	32	A shortcut straight across the ring: the farthest pair (distance 5) becomes distance 1.
10 portals, `1 6` twice	32	A duplicate line just adds the same connection twice. BFS doesn't care.
6 portals, `1 4`, `2 5`, `3 6`	12	Strain falls from 54 to 42: every portal is now at most 2 teleports from every other one.

Shared approach: the graph and measuring strain

All three parts work the same way; only the ring itself differs (1 neighbour each side in Part I, 3 in Parts II and III).

- **The graph** is an adjacency list: for every portal, the list of portals it connects to. Every connection is added in both directions.
- **Route sum from one portal:** a BFS from that portal. Each portal reached for the first time is one step further than the portal it was reached from, and its distance is added to the sum straight away, so no second pass is needed.
- **Strain:** run that BFS from every portal and add up the route sums.

Cost: one BFS visits every portal and connection once, so it's O(N + E). The strain runs N of them: O(N · (N + E)), which is $O(N^2)$ here because each portal has only a handful of connections.

PART I The plain ring

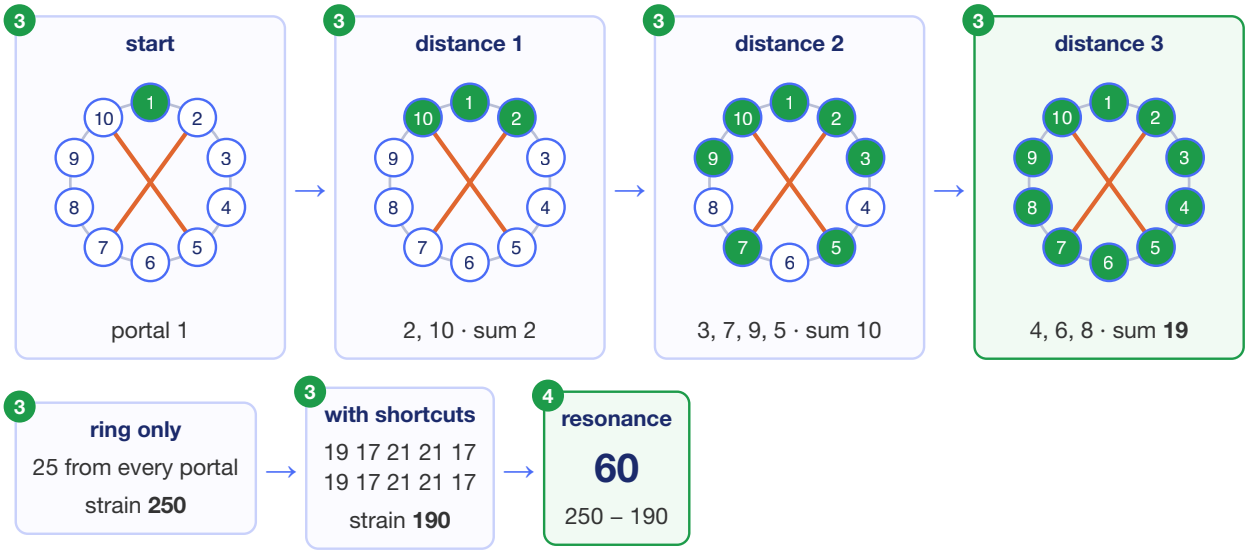
Each portal is linked to the one before and the one after it (and N wraps round to 1). Compute the strain without and with the listed shortcuts and print the difference.

Step-by-step solution

1. Read N and the list of shortcuts.
2. Build the ring as an adjacency list: link every portal to the next one, with N wrapping round to 1.
3. For each start portal, run a BFS and add up the distances to all other portals. The total over all starts is the strain.
4. Do this once for the ring alone and once with the shortcuts added, and print the difference.

How my solution works

Link every portal to the next (N back to 1), run a **BFS** from every portal, and add up the distances. BFS from portal 1 with the sample's shortcuts, one distance layer at a time:



Dots match the step numbers above. Green = reached so far. The route sums match the puzzle exactly.

PART II A wider ring and one crystal pair

Now every portal is linked to the **3** nearest portals on each side. The listed shortcuts are there too, and you may add one more two-way connection between **any** two portals. Choose it to make the resonance as large as possible. The "before" strain is still the ring alone.

Worked examples

Configuration	Route sums from portals 1 … 10	Strain
ring only	12 each	120
ring + 5–10 + 2–7		116
… + crystals 1–5	11 11 12 12 10 12 11 12 12 11	114

Resonance 120 − 114 = **6**. 13 of the 45 pairs reach 114 (e.g. 1–5, 2–6, 3–7); the other 32 don't help, so there are "several equally good choices".

Extra examples of my own (copies of my Part II solution)

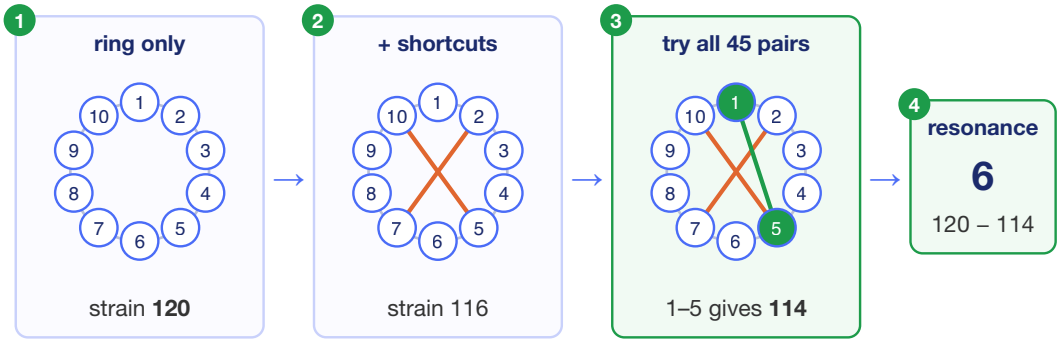
Notes	Answer	Why
10 portals, no shortcuts	2	120 → 118
20 portals, `5 10`, `2 7`	64	a bigger ring saves far more
7 portals, no shortcuts	0	everyone is already 1 teleport apart

Step-by-step solution

1. Build the ring with links to the 3 nearest portals on each side, and measure its strain ("before").
2. Add the listed shortcuts.
3. For every pair of portals a < b: add the connection a–b, measure the strain, keep the smallest, then remove the connection again.
4. Print "before" minus the smallest strain.

How my solution works

Measure the ring alone, then the ring with its shortcuts, then try **every pair** as the crystal connection: add it, measure, keep the best, and take it off again (it was the last thing added, so the graph never needs rebuilding).

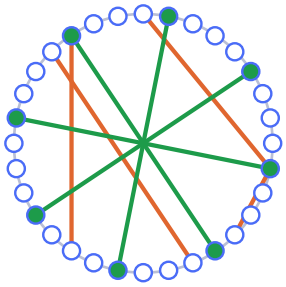


Dots match the step numbers above. (Each portal also links to its 3 nearest on each side; the circle stands for those links.)

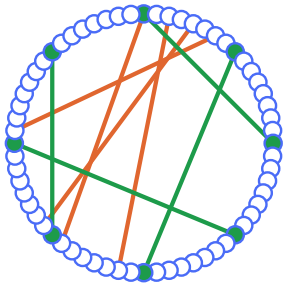
PART III Eight crystals, evenly spaced

Same ring as Part II, with its listed shortcuts. Now 8 crystals must sit at **equal intervals**: if the first is at portal *s*, the others are at *s* + g, *s* + 2g, …, with g = N / 8. Then pair the 8 crystals into 4 two-way connections. Choose the start *s* and the pairing to maximise the resonance.

Worked examples



32 portals: crystals 2, 6, … , 30



64 portals: crystals 1, 9, … , 57

Red: listed shortcuts. Green: the crystal connections my solution found first (filled dots are crystals).

Sample	g	Ring only	+ shortcuts	Best with crystals	Resonance
32 portals	4	3072	2610	2218	**854**
64 portals (sample3.txt)	8	23232	16994	13598	**9634**

The 32-portal example (not in the sample file, run through a copy of my solution) gives 854 with exactly the puzzle's placement. Neither best pairing is simply "opposite pairs".

Best strain for each start position (instrumented copy)

Start *s*	1	2	3	4	5	6	7	8
32 portals	2226	**2218**	2228	2228				
64 portals	**13598**	13606	13626	13682	13678	13736	13630	13670

The start matters because the shortcuts make the ring uneven; start g + 1 would repeat start 1, so the search stops at g.

Step-by-step solution

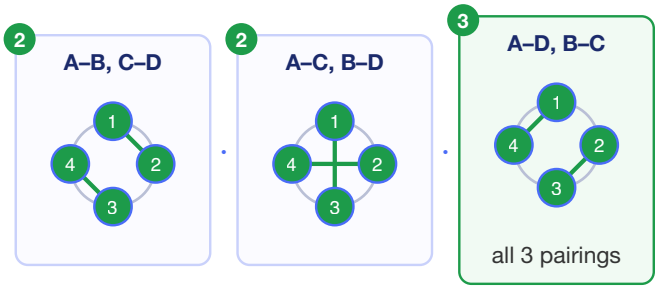
1. Build the Part II ring with its shortcuts, and measure the ring-only strain.
2. Work out the gap g = N / 8. For each start s = 1 … g, the crystals are s, s + g, …, s + 7g.
3. Recursively pair the crystals: take the first unpaired one, pair it with each remaining one in turn, and recurse on the rest.
4. When all 4 connections are in place, measure the strain and keep the smallest. Undo each connection on the way back.
5. Print the ring-only strain minus the smallest strain.

How my solution works

Everything up to the search is Part II. For each start s = 1 … g (g = N / 8), the crystals are s, s + g, …, s + 7g; a **recursive pairing** step tries every way to pair them up (add a connection, recurse, undo), measuring the strain once all 4 connections are in place.

How the pairing recursion lists every pairing once

Always pair the **first** unpaired crystal with each of the others in turn, then recurse on the rest. For 4 crystals:



For 8 crystals that's 7 × 5 × 3 × 1 = **105** pairings per start: 420 strains for 32 portals, 840 for 64. Overall about 13 · N^3 work, cheaper than Part II's N^4 for large N; the real input runs in about 0.2 s.

> **Fragile: N must be a multiple of 8.** The gap N / 8 is not rounded. For a 30-portal ring, g = 3.75, the crystals land on "portals" like 4.75, which don't exist: a copy of my solution crashed when it tried to connect one. The puzzle's "perfectly equal intervals" only makes sense when 8 divides N, and both examples (and the real input) satisfy that, so this is fine for the quest, just worth knowing.
> **Also harmless:** starting from the "no crystals" strain, and duplicate connections (a crystal pair matching an existing link).

Summary

Part	Ring links per side	Extra connections you choose	Search	Strains measured (sample)	Sample answer
I	1	none	two strains	2	60
II	3	1, anywhere	all N(N – 1)/2 pairs	47	6
III	3	4, from 8 evenly spaced crystals	g starts × 105 pairings	842	9634

Key ideas: in an unweighted graph, **BFS** from every node gives all shortest distances (all-pairs shortest paths). When the choices are few enough, simply **try them all**: add an edge, measure, and undo it (backtracking), instead of rebuilding the graph. For pairings, always pair the first remaining item to list every perfect matching exactly once.